

DSA0602-Class Activity 2

Scenario

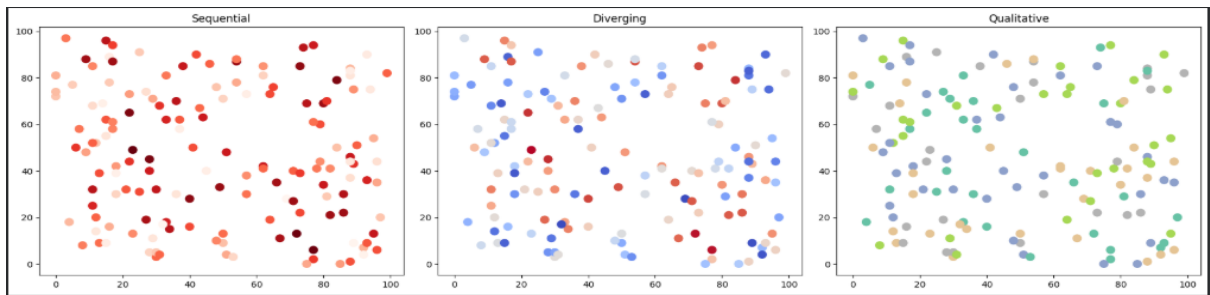
1. Smart City Air Quality Monitoring

Python code :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
np.random.seed(10)
n = 150
df = pd.DataFrame({
    "X": np.random.randint(0,100,n),
    "Y": np.random.randint(0,100,n),
    "PM25": np.random.randint(20,300,n),
    "NO2": np.random.randint(10,120,n),
    "CO": np.random.uniform(0.2,4,n),
    "Temp": np.random.randint(20,40,n),
    "Humidity": np.random.randint(30,90,n)
})
fig,ax=plt.subplots(1,3,figsize=(18,5))
ax[0].scatter(df.X,df.Y,c=df.PM25,cmap='Reds',s=80)
ax[0].set_title("Sequential")
ax[1].scatter(df.X,df.Y,c=df.PM25-150,cmap='coolwarm',s=80)
ax[1].set_title("Diverging")
ax[2].scatter(df.X,df.Y,c=np.random.randint(0,5,n),cmap='Set2',s=80)
ax[2].set_title("Qualitative")
plt.tight_layout()
```

Scenario

```
plt.show()
```



R-program

```
library(ggplot2)

set.seed(10)

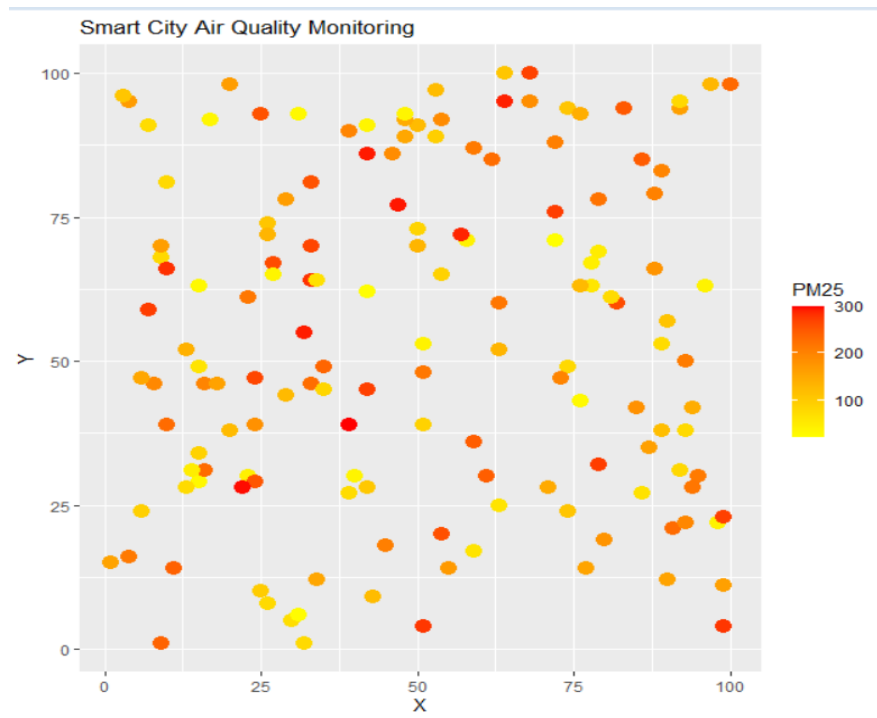
n <- 150

df <- data.frame(
  X = sample(1:100, n, replace = TRUE),
  Y = sample(1:100, n, replace = TRUE),
  PM25 = sample(20:300, n, replace = TRUE)
)

ggplot(df, aes(X, Y, color = PM25)) +
  geom_point(size = 4) +
  scale_color_gradient(low = "yellow", high = "red") +
  ggtitle("Smart City Air Quality Monitoring")
```

Output Screenshot

Scenario



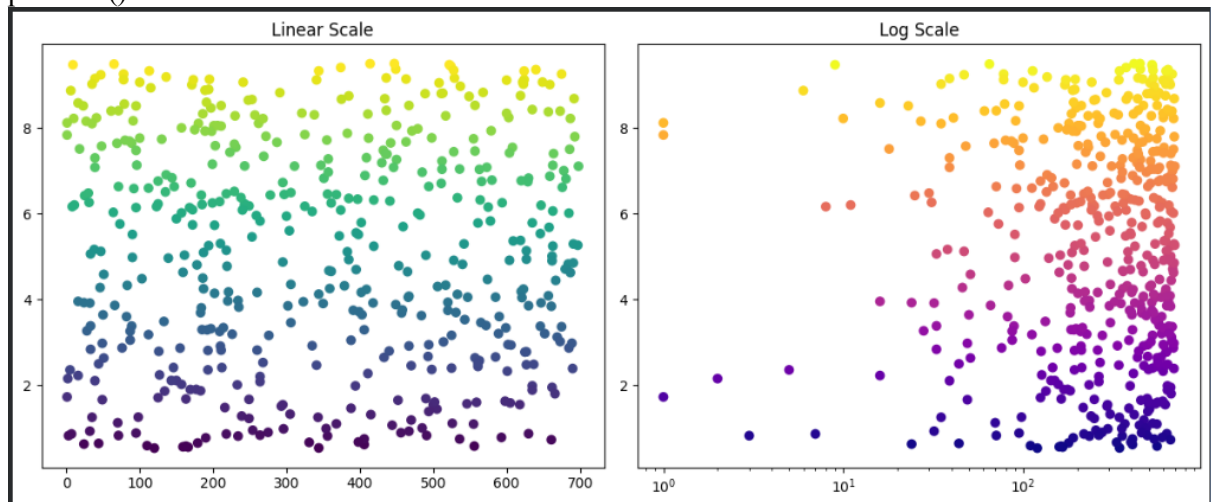
2. Earthquake Risk Analysis

Python code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
np.random.seed(20)
n=500
df=pd.DataFrame({
    "Magnitude":np.random.uniform(0.5,9.5,n),
    "Depth":np.random.randint(1,700,n)
})
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.scatter(df.Depth,df.Magnitude,c=df.Magnitude,cmap='viridis')
plt.title("Linear Scale")
```

Scenario

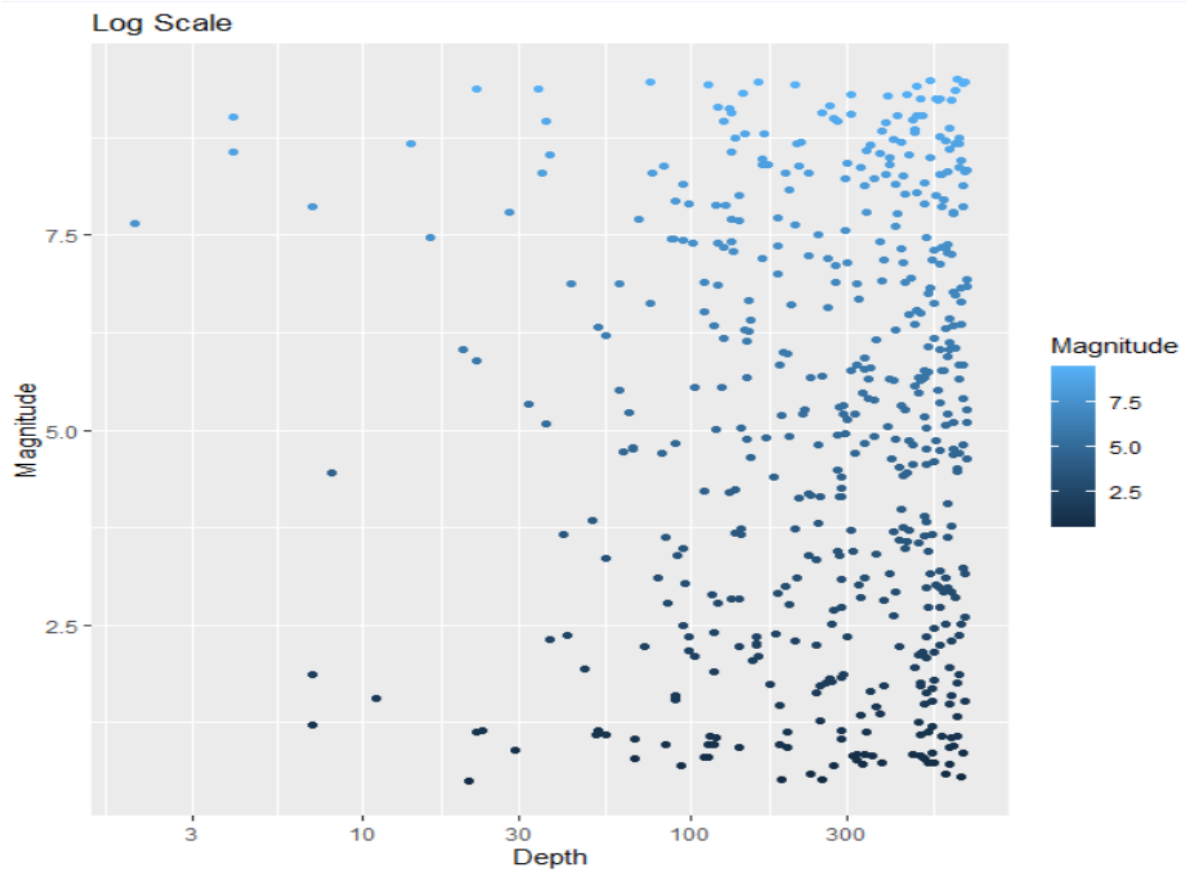
```
plt.subplot(1,2,2)
plt.scatter(df.Depth,df.Magnitude,c=df.Magnitude,cmap='plasma')
plt.xscale('log')
plt.title("Log Scale")
plt.tight_layout()
plt.show()
```



R-Program:

```
library(ggplot2)
set.seed(20)
df <- data.frame(
  Magnitude = runif(500, 0.5, 9.5),
  Depth = sample(1:700, 500, replace = TRUE)
)
ggplot(df, aes(Depth, Magnitude, color = Magnitude)) +
  geom_point() +
  ggtitle("Linear Scale")
ggplot(df, aes(Depth, Magnitude, color = Magnitude)) +
  geom_point() +
  scale_x_log10() +
  ggtitle("Log Scale")
```

Scenario



3. Wind Direction Analysis

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(30)

direction=np.random.randint(0,360,365)
speed=np.random.uniform(5,40,365)

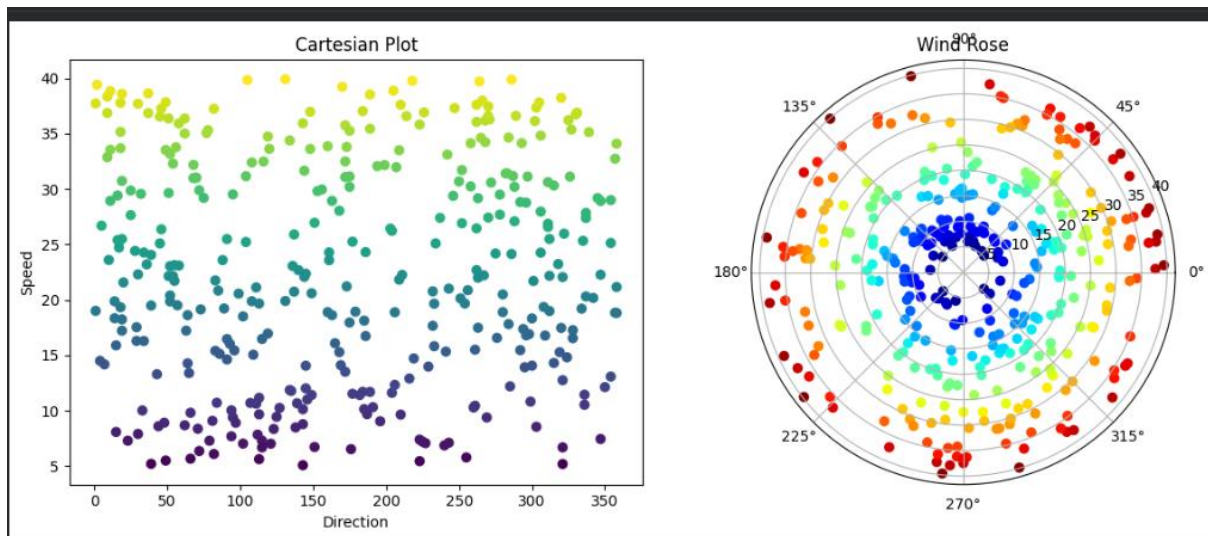
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)

plt.scatter(direction,speed,c=speed,cmap='viridis')
plt.xlabel("Direction")
plt.ylabel("Speed")
plt.title("Cartesian Plot")

ax=plt.subplot(1,2,2,projection='polar')
```

Scenario

```
theta=np.deg2rad(direction)
ax.scatter(theta,speed,c=speed,cmap='jet')
ax.set_title("Wind Rose")
plt.tight_layout()
plt.show()
```

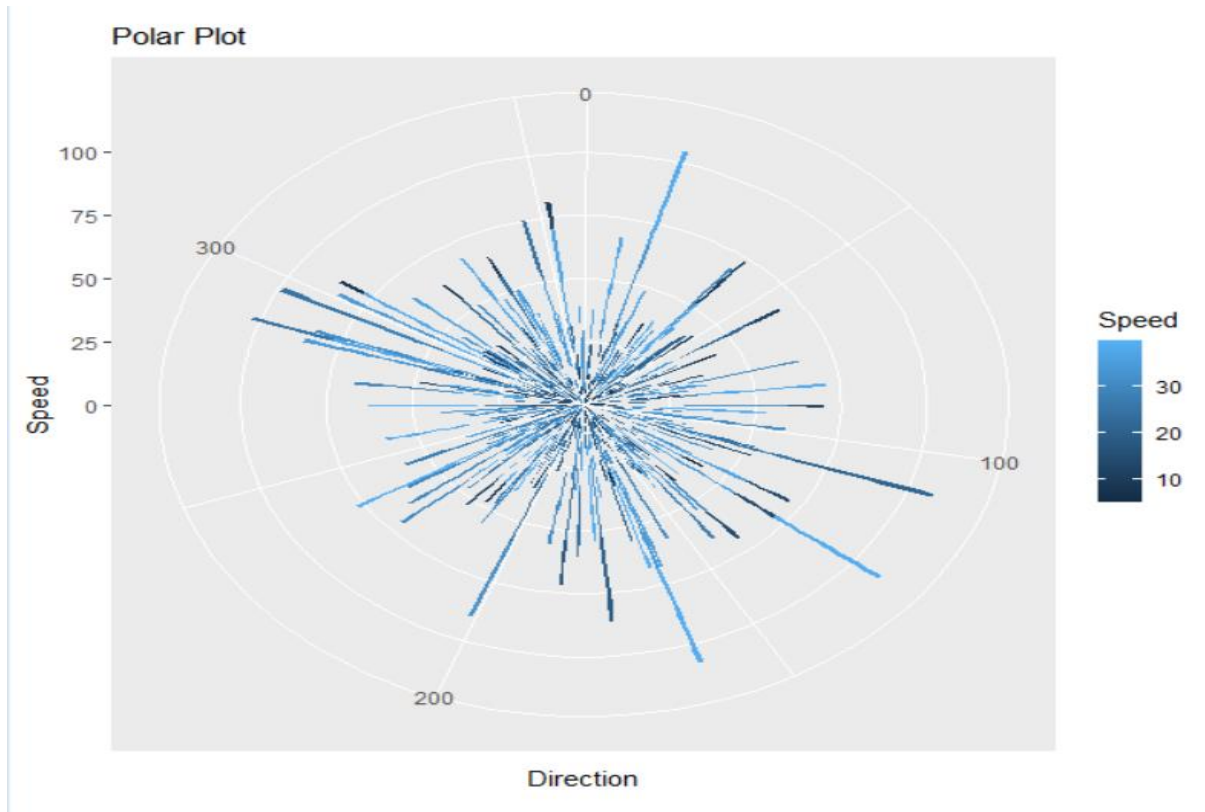


R-program

```
library(ggplot2)
set.seed(30)
df <- data.frame(
  Direction = sample(0:359, 365, replace = TRUE),
  Speed = runif(365, 5, 40)
)
ggplot(df, aes(Direction, Speed, color = Speed)) +
  geom_point() +
  ggtitle("Cartesian Plot")
ggplot(df, aes(Direction, Speed, fill = Speed)) +
  geom_bar(stat = "identity") +
  coord_polar() +
```

Scenario

ggtitle("Polar Plot")



4. Hospital Patient Monitoring Dashboard

Python code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(40)

time=np.arange(100)

heart=np.random.randint(60,130,100)
oxygen=np.random.randint(85,100,100)

plt.figure(figsize=(12,5))

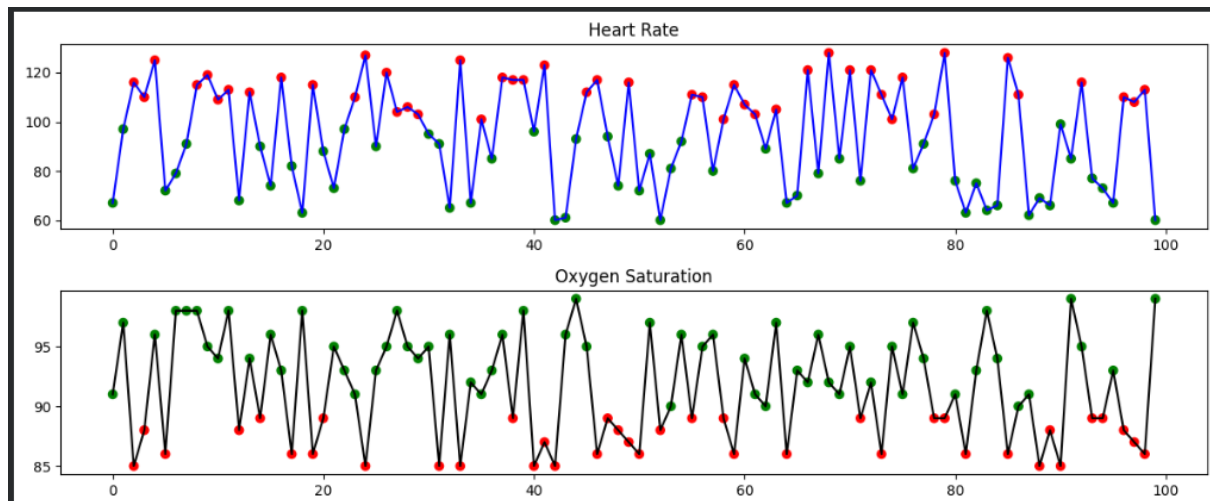
plt.subplot(2,1,1)
plt.plot(time,heart,color='blue')
plt.scatter(time,heart,c=np.where(heart>100,'red','green'))
plt.title("Heart Rate")

plt.subplot(2,1,2)
```


Scenario

```
plt.plot(time,oxygen,color='black')
plt.scatter(time,oxygen,c=np.where(oxygen<90,'red','green'))
plt.title("Oxygen Saturation")

plt.tight_layout()
plt.show()
```

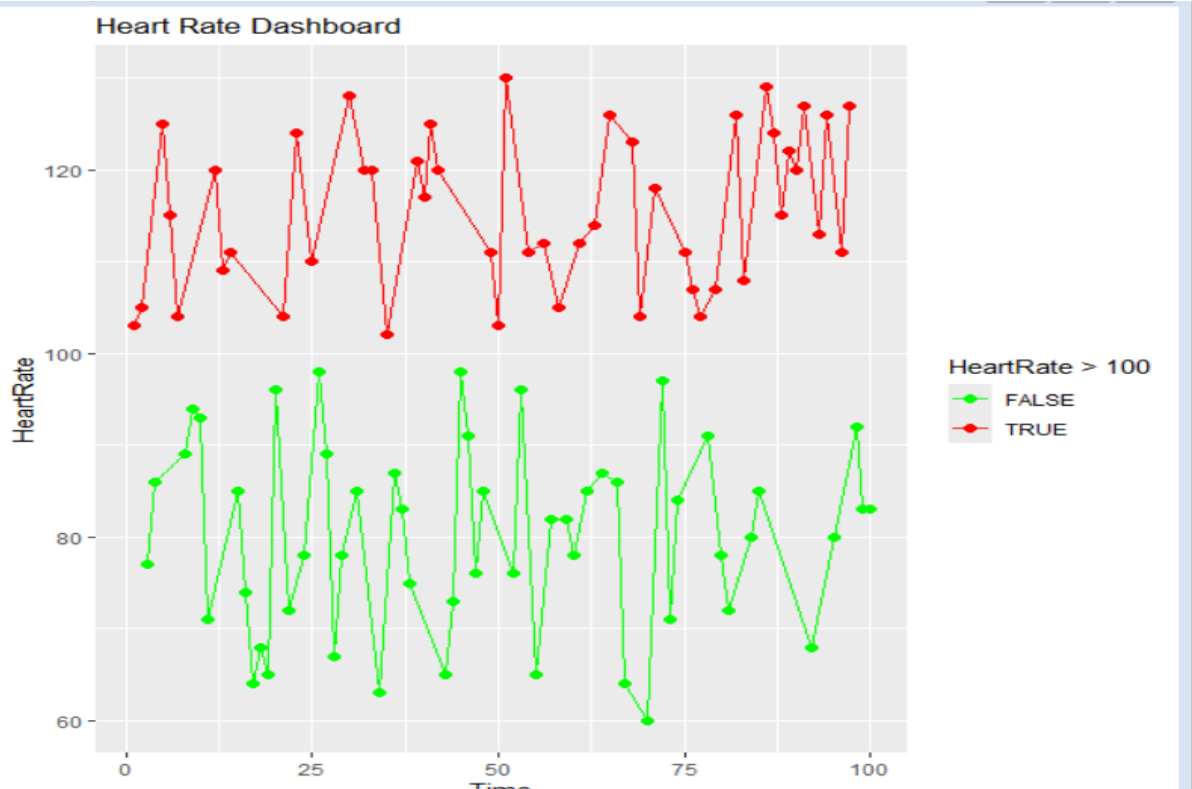


R-program

```
library(ggplot2)
set.seed(40)
time <- 1:100
heart <- sample(60:130, 100, replace = TRUE)

df <- data.frame(Time = time, HeartRate = heart)

ggplot(df, aes(Time, HeartRate, color = HeartRate > 100)) +
  geom_line() +
  geom_point(size = 2) +
  scale_color_manual(values = c("green", "red")) +
  ggtitle("Heart Rate Dashboard")
```



5. Global Climate Change Study

Python code:

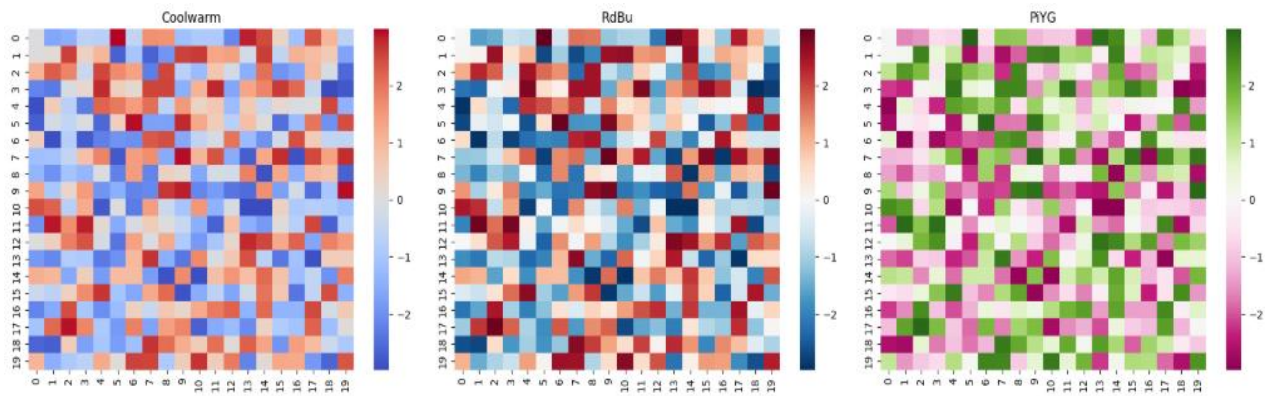
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
np.random.seed(50)
data=np.random.uniform(-3,3,(20,20))
fig,ax=plt.subplots(1,3,figsize=(18,5))
sns.heatmap(data,cmap='coolwarm',ax=ax[0])
ax[0].set_title("Coolwarm")
sns.heatmap(data,cmap='RdBu_r',ax=ax[1])
ax[1].set_title("RdBu")
sns.heatmap(data,cmap='PiYG',ax=ax[2])
```

Scenario

```
ax[2].set_title("PiYG")
```

```
plt.tight_layout()
```

```
plt.show()
```



R-program

```
library(ggplot2)
```

```
library(reshape2)
```

```
set.seed(50)
```

```
data <- matrix(runif(400, -3, 3), nrow = 20)
```

```
df <- melt(data)
```

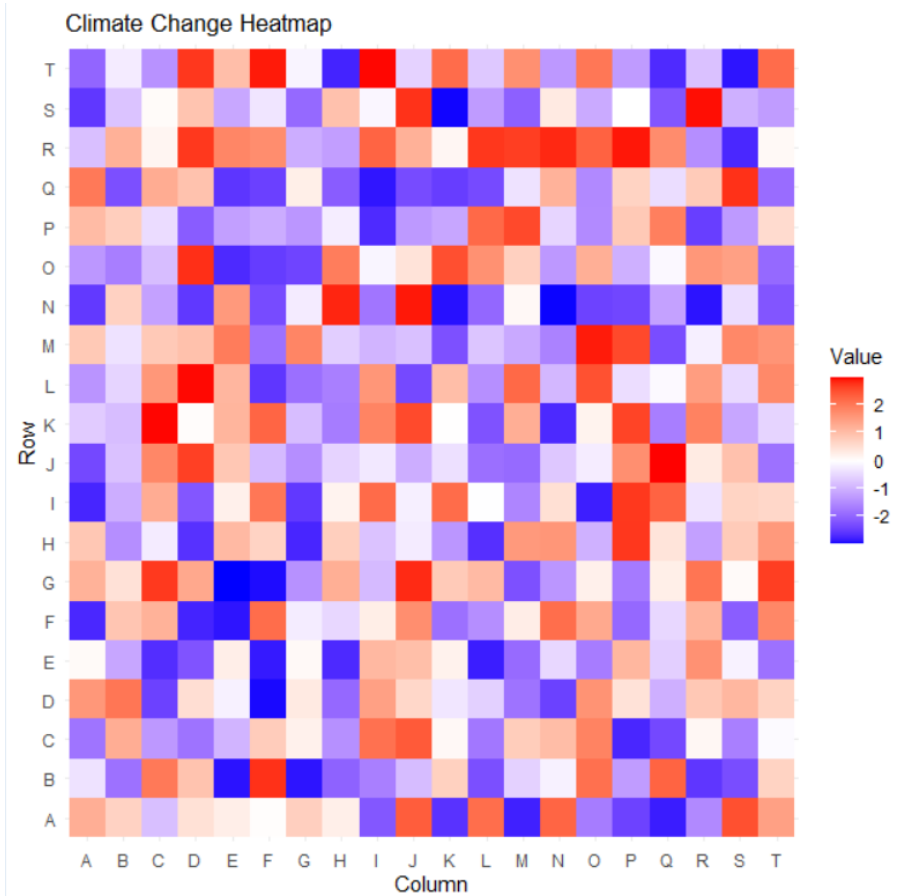
```
ggplot(df, aes(Var1, Var2, fill = value)) +
```

```
  geom_tile() +
```

```
  scale_fill_gradient2(low = "blue", mid = "white", high = "red") +
```

```
  ggtitle("Climate Change Heatmap")
```

Scenario



6. E-Commerce Sales Analytics

Python code:\

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(60)

categories=[f"C{i}" for i in range(1,11)]

sales=np.random.randint(1000,10000,10)

plt.figure(figsize=(10,5))

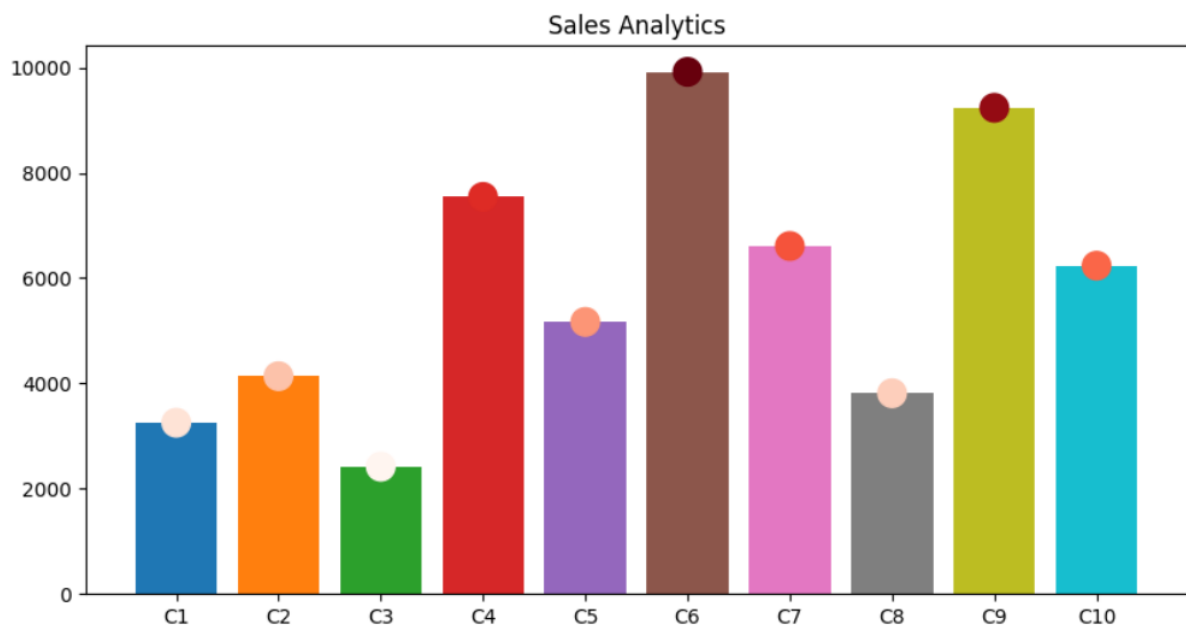
plt.bar(categories,sales,color=plt.cm.tab10(range(10)))

plt.scatter(categories,sales,c=sales,cmap='Reds',s=200)

plt.title("Sales Analytics")

plt.show()
```

Scenario



R-program:

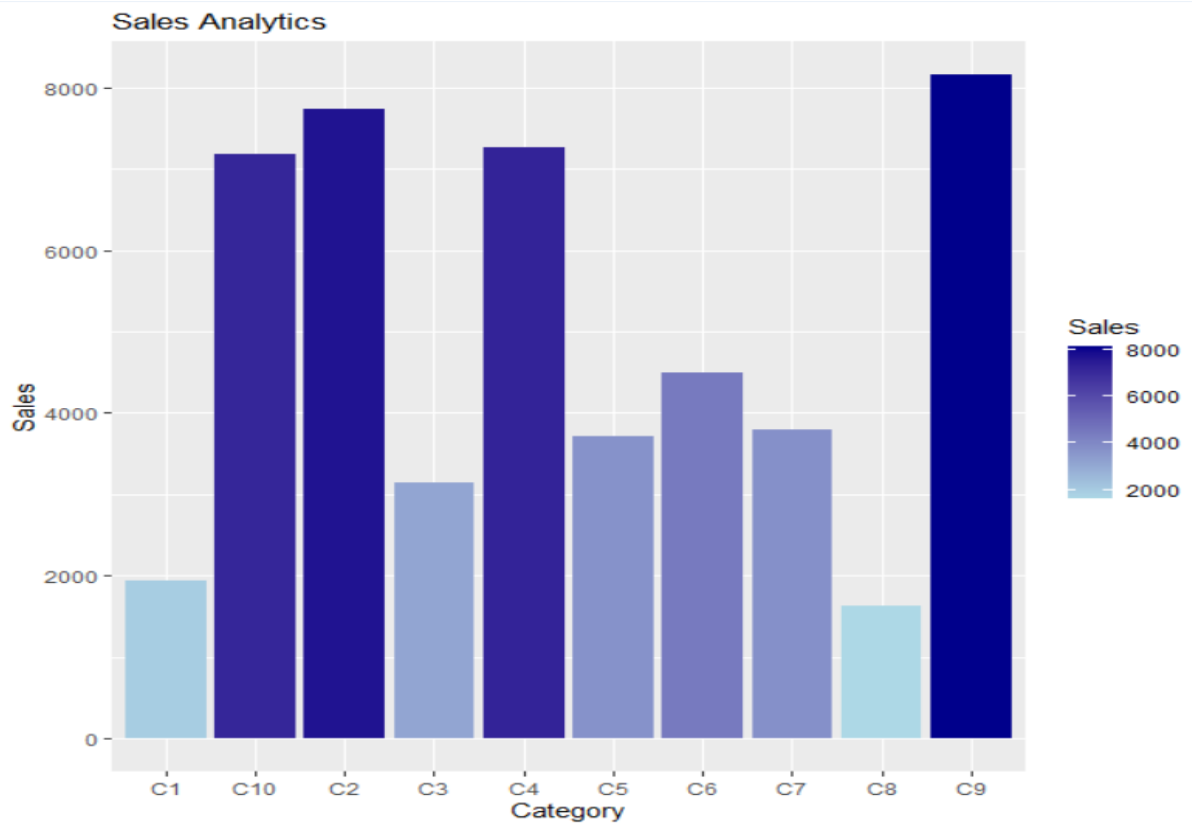
```
library(ggplot2)
```

```
set.seed(60)
```

```
df <- data.frame(  
  Category = paste0("C",1:10),  
  Sales = sample(1000:10000,10)  
)
```

```
ggplot(df,aes(Category,Sales,fill=Sales))+  
  geom_bar(stat="identity")+  
  scale_fill_gradient(low="lightblue",high="darkblue")+  
  ggtitle("Sales Analytics")
```

Scenario

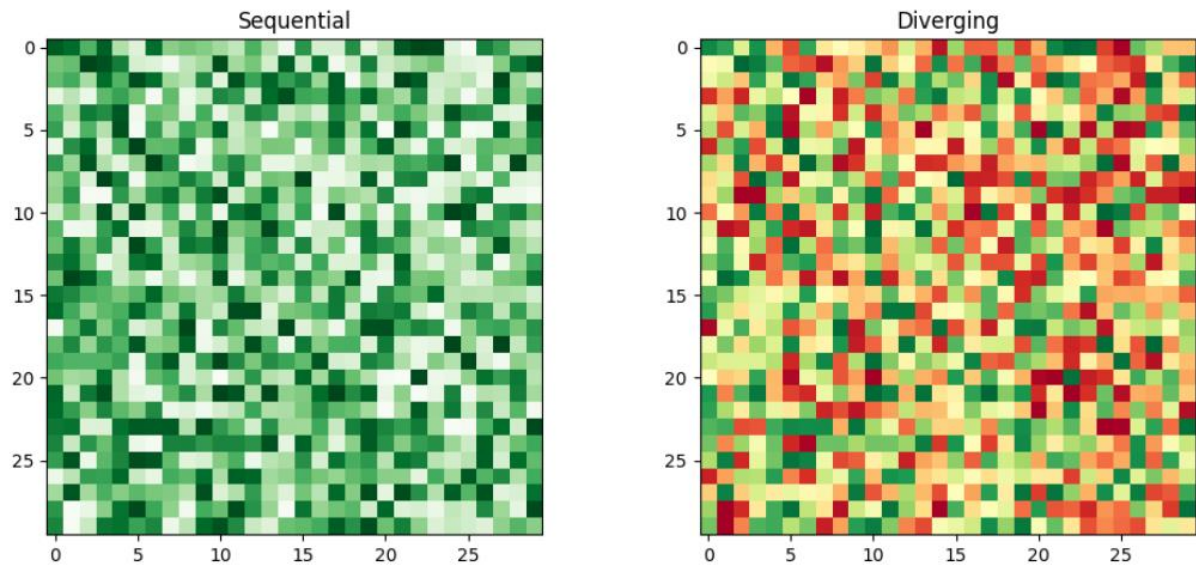


7. Satellite Image Vegetation Analysis

Python code:

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(70)
ndvi=np.random.uniform(-1,1,(30,30))
fig,ax=plt.subplots(1,2,figsize=(12,5))
ax[0].imshow(ndvi,cmap='Greens')
ax[0].set_title("Sequential")
ax[1].imshow(ndvi,cmap='RdYlGn')
ax[1].set_title("Diverging")
plt.show()
```

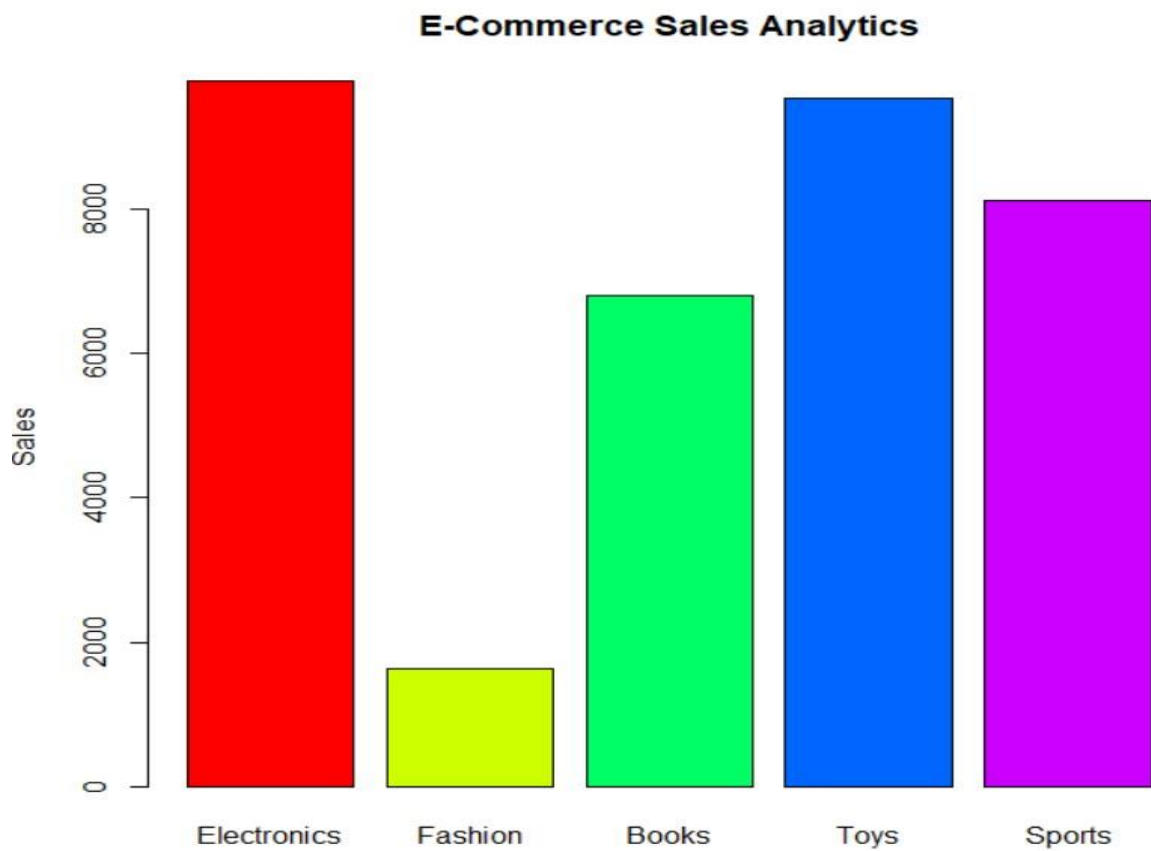
Scenario



R-program

```
library(ggplot2)
library(reshape2)
set.seed(70)
mat <- matrix(runif(900,-1,1),30)
df <- melt(mat)
ggplot(df,aes(Var1,Var2,fill=value))+
geom_tile()+
scale_fill_gradient2(low="brown",mid="yellow",high="green")+
ggtitle("NDVI Vegetation Analysis")
```

Scenario



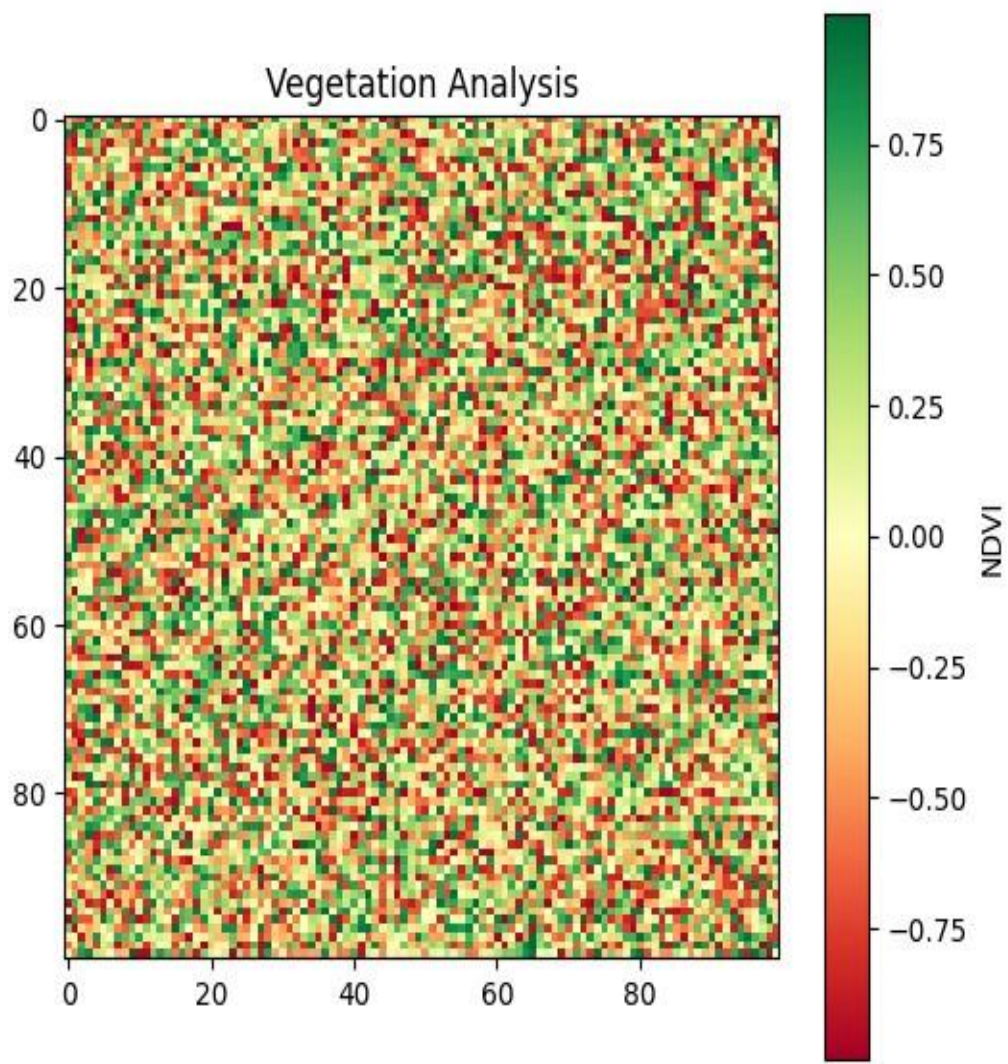
7: Satellite Image Vegetation Analysis

CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt
ndvi=np.random.uniform(-1,1,(100,100))
plt.figure(figsize=(6,6))
plt.imshow(ndvi,cmap="RdYlGn")
plt.colorbar(label="NDVI")
plt.title("Vegetation Analysis")
plt.show()
```

OUTPUT:

Scenario



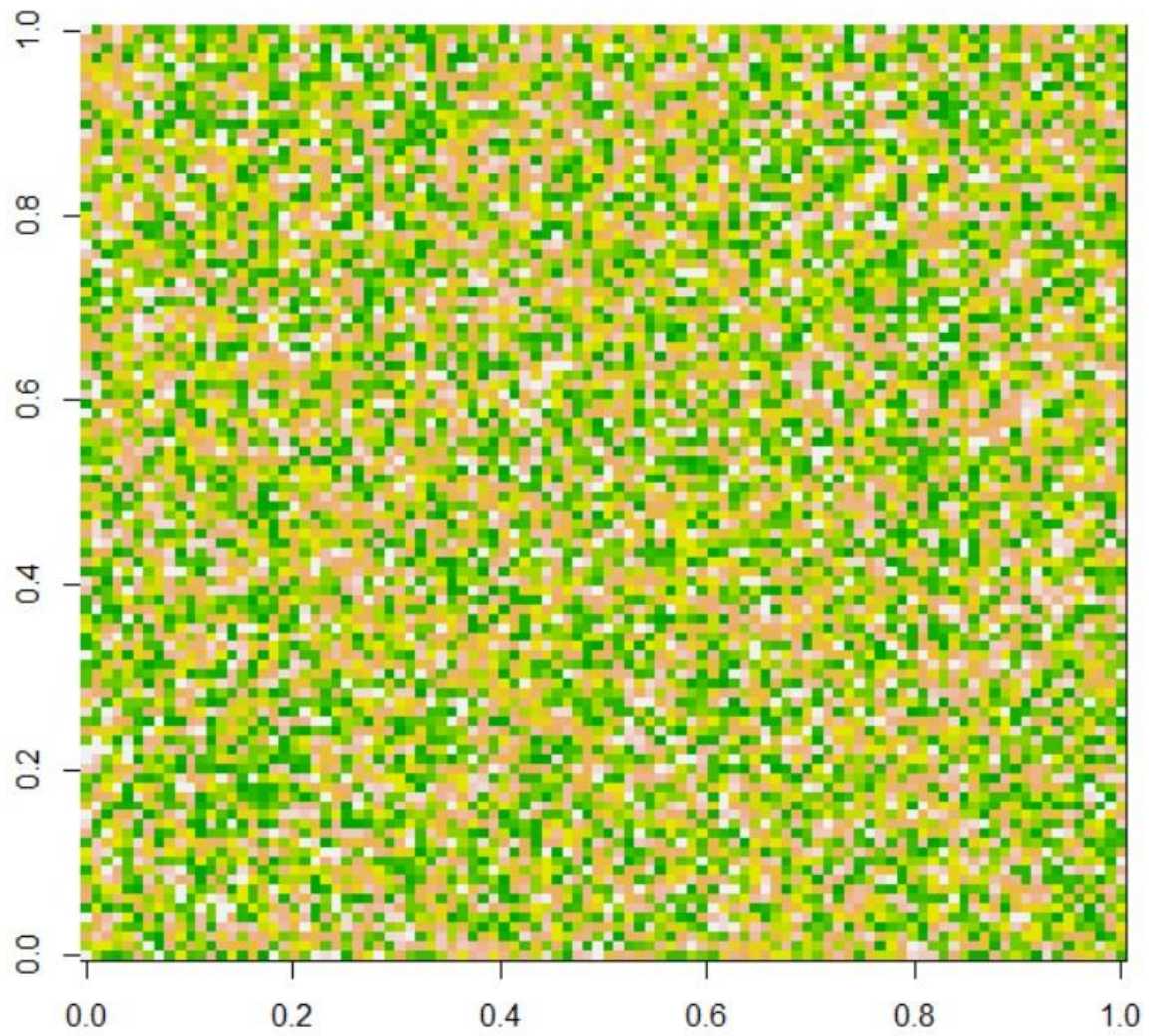
CODE: R program

```
set.seed(7)
ndvi      <-      matrix(runif(10000,-
1,1),100,100)
image(ndvi,      col=terrain.colors(20),
      main="Vegetation Analysis (NDVI)")
```

OUTPUT:

Scenario

Vegetation Analysis (NDVI)

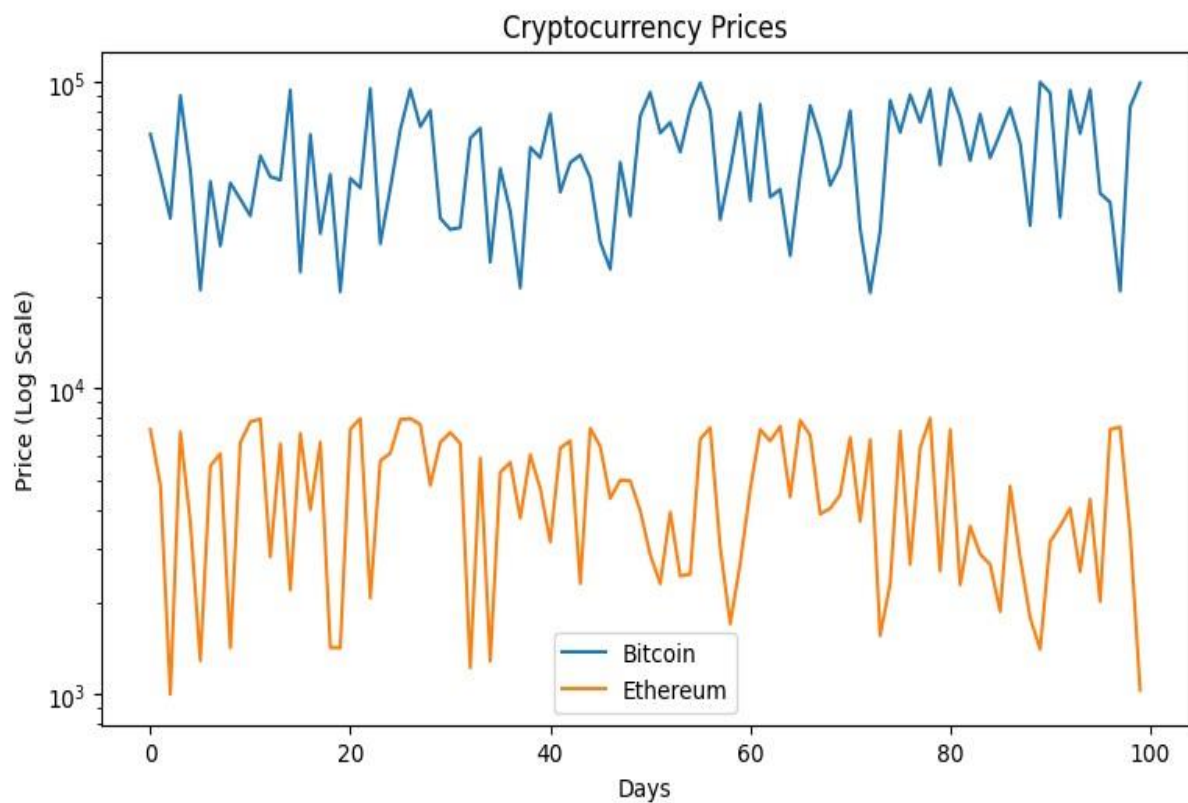


Scenario

8: Cryptocurrency Market Volatility

```
import numpy as np
import matplotlib.pyplot as plt
days=np.arange(100)
btc=np.random.uniform(20000,100000,100)
eth=np.random.uniform(1000,8000,100)
plt.figure(figsize=(9,5))
plt.plot(days,btc,label="Bitcoin")
plt.plot(days,eth,label="Ethereum")
plt.yscale("log")
plt.legend()
plt.title("Cryptocurrency Prices")
plt.xlabel("Days")
plt.ylabel("Price (Log Scale)")
plt.show()
```

OUTPUT:

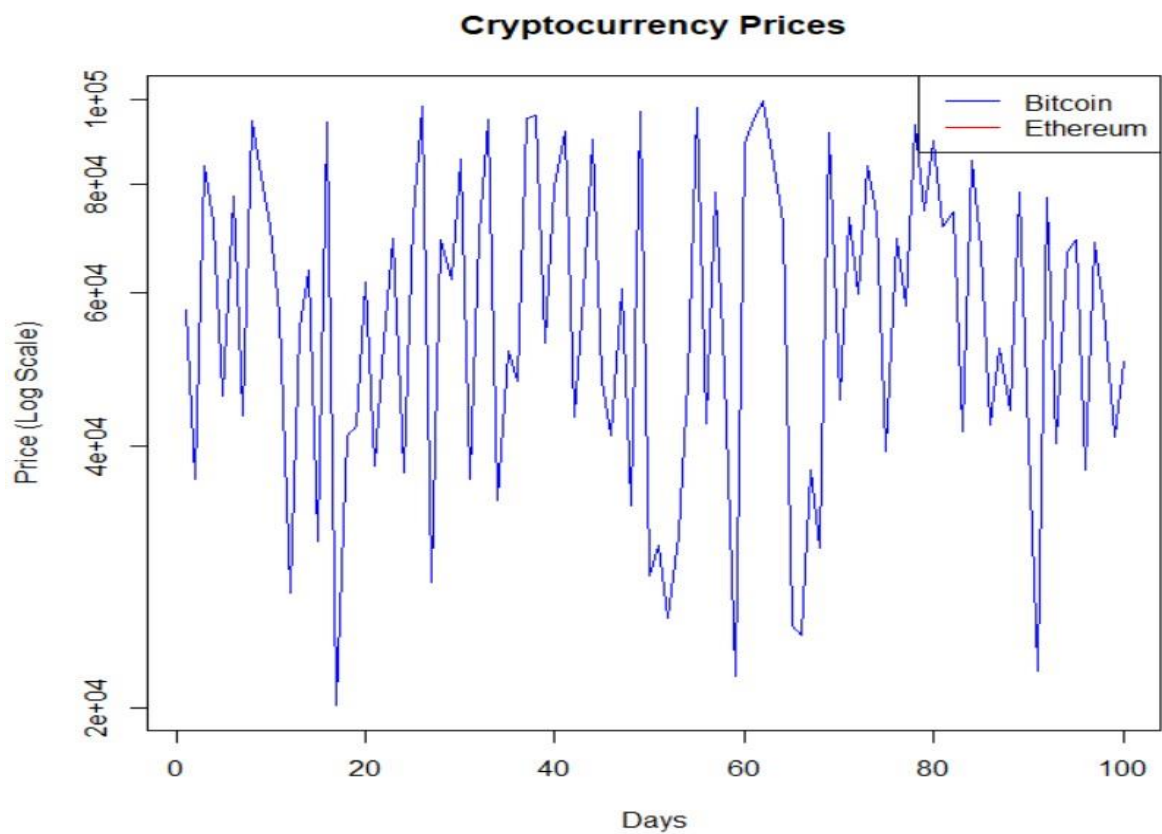


CODE: Rprogram

Scenario

```
set.seed(8)
days <- 1:100 btc <-
runif(100,20000,100000) eth <-
runif(100,1000,8000)
plot(days, btc, type="l",
log="y", col="blue",
xlab="Days", ylab="Price (Log
Scale)", main="Cryptocurrency
Prices")
lines(days, eth,
col="red")
legend("topright",
legend=c("Bitcoin","Ethereum"),
col=c("blue","red"), lty=1)
```

OUTPUT:

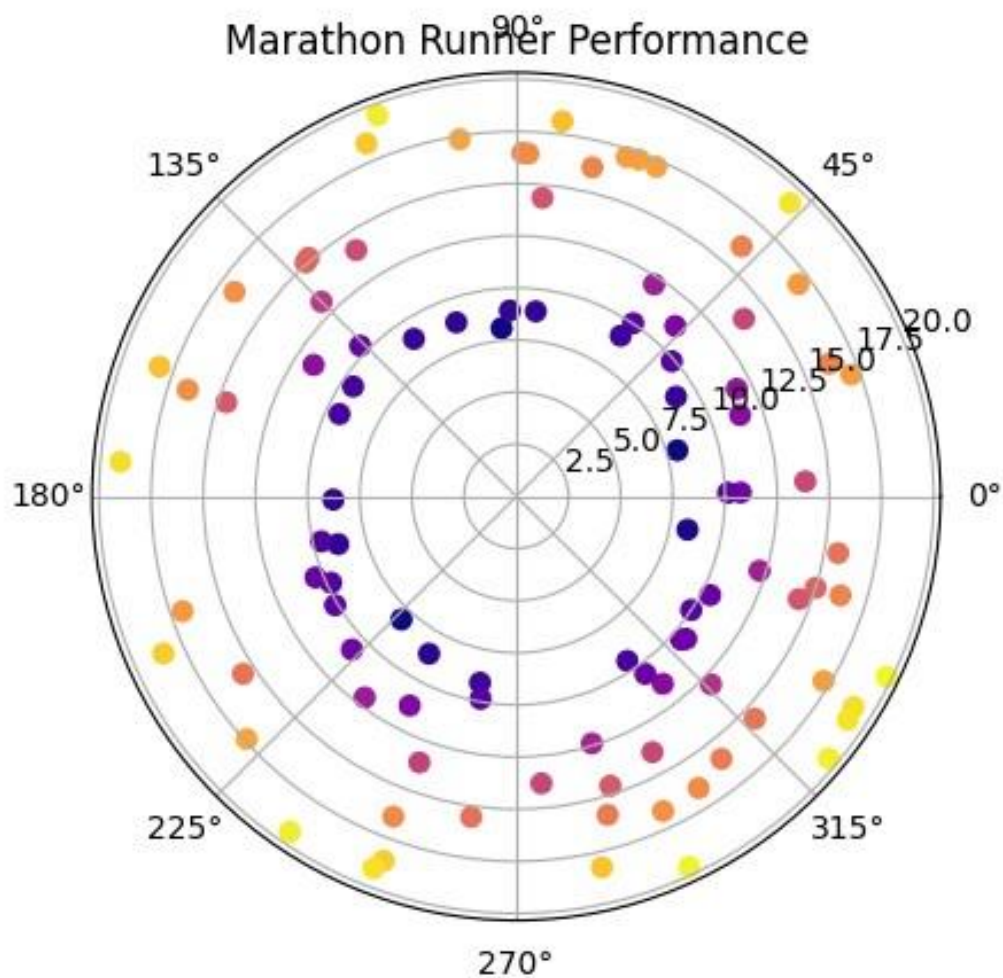


Scenario

9: Marathon Runner Performance Analysis

```
import numpy as np
import matplotlib.pyplot as plt
direction=np.deg2rad(np.random.randint(0,360,100))
speed=np.random.uniform(8,20,100)
ax=plt.subplot(111,projection="polar")
ax.scatter(direction,speed,c=speed,cmap="plasma")
plt.title("Marathon Runner Performance")
plt.show()
```

OUTPUT:

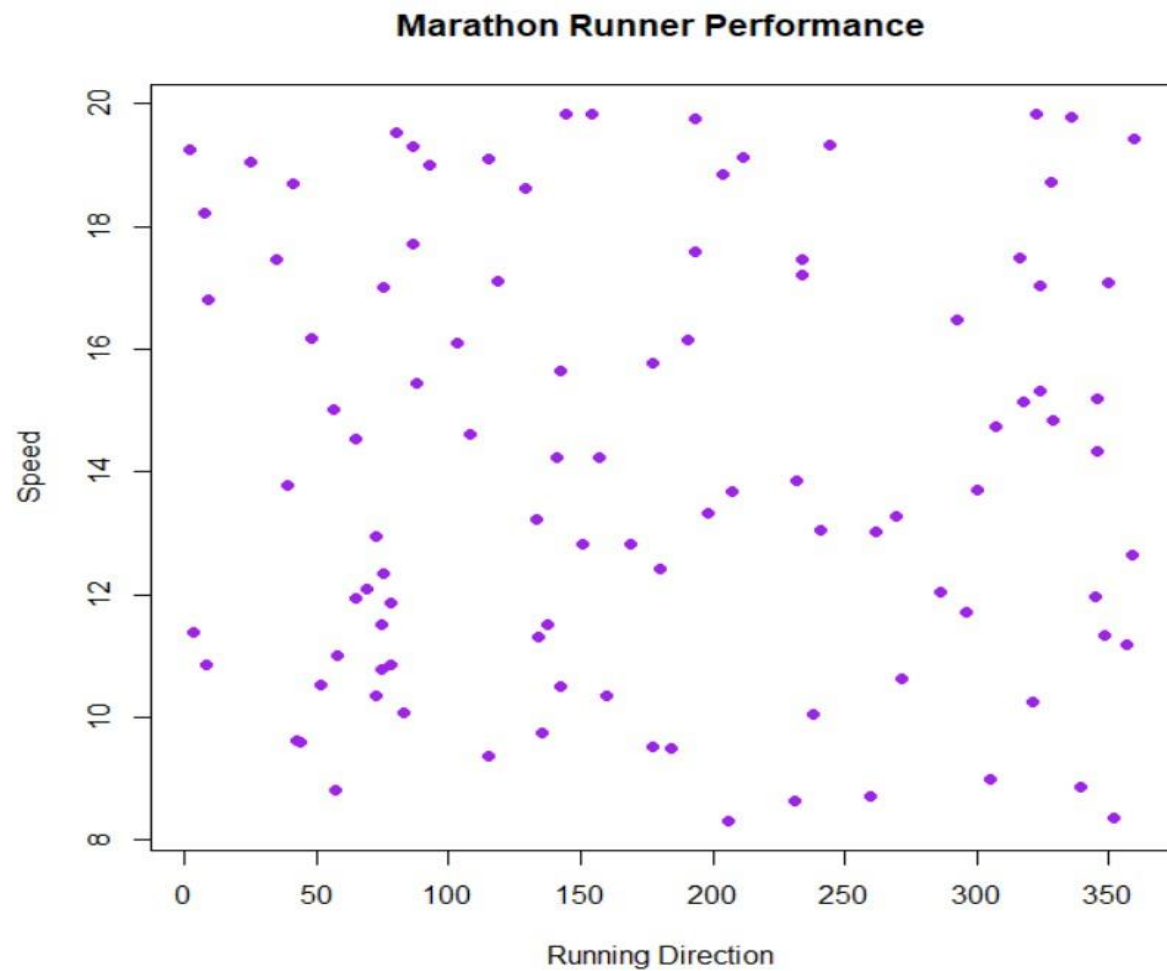


CODE: R Program

Scenario

```
set.seed(9)
direction <-
runif(100,0,360) speed <-
runif(100,8,20)
plot(direction,      speed,
pch=19,      col="purple",
xlab="Running Direction",
ylab="Speed",      main="Marathon Runner
Performance")
```

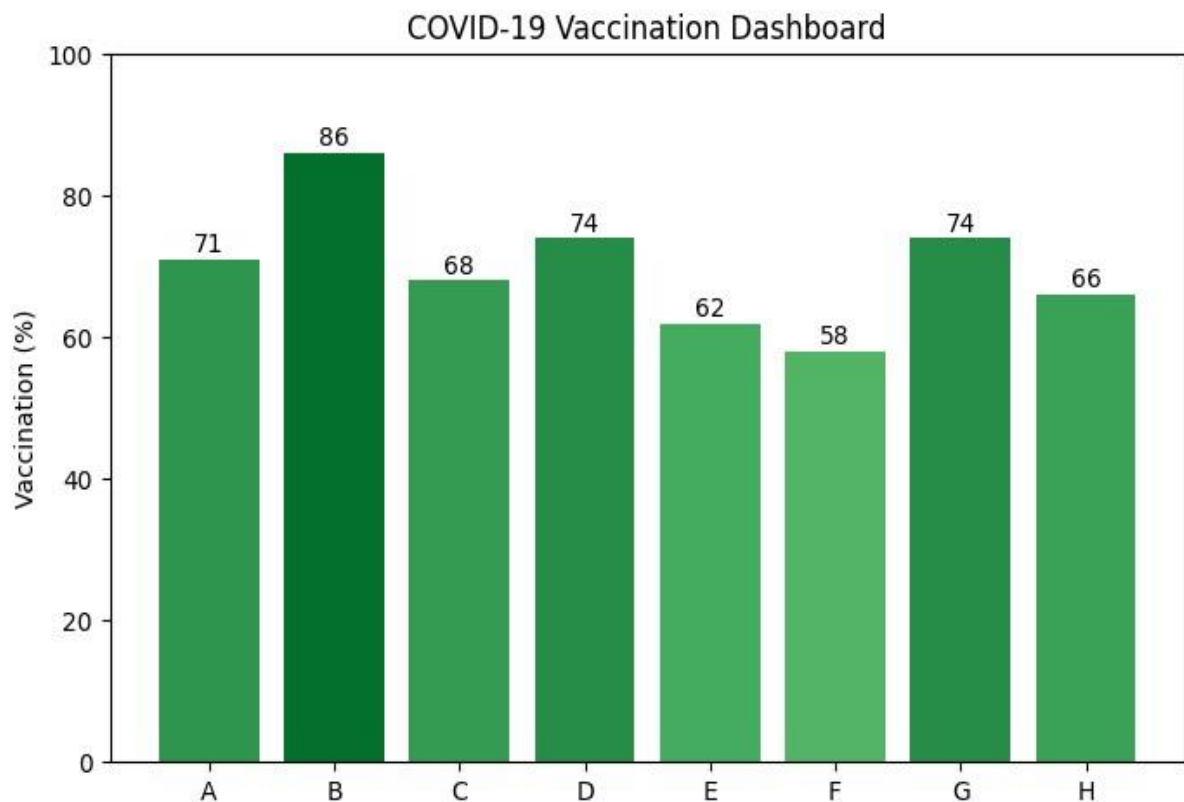
OUTPUT:



Scenario

```
import numpy as np
import matplotlib.pyplot as plt
districts=["A","B","C","D","E","F","G","H"]
vaccinated=np.random.randint(50,100,8)
plt.figure(figsize=(8,5))
bars=plt.bar(districts,vaccinated,color=plt.cm.Greens(vaccinated/100))
plt.ylabel("Vaccination (%)")
plt.title("COVID-19 Vaccination Dashboard")
plt.ylim(0,100)
for bar,val in zip(bars,vaccinated):
    plt.text(bar.get_x()+bar.get_width()/2,val+1,str(val),
             ha='center')
plt.show()
```

OUTPUT:

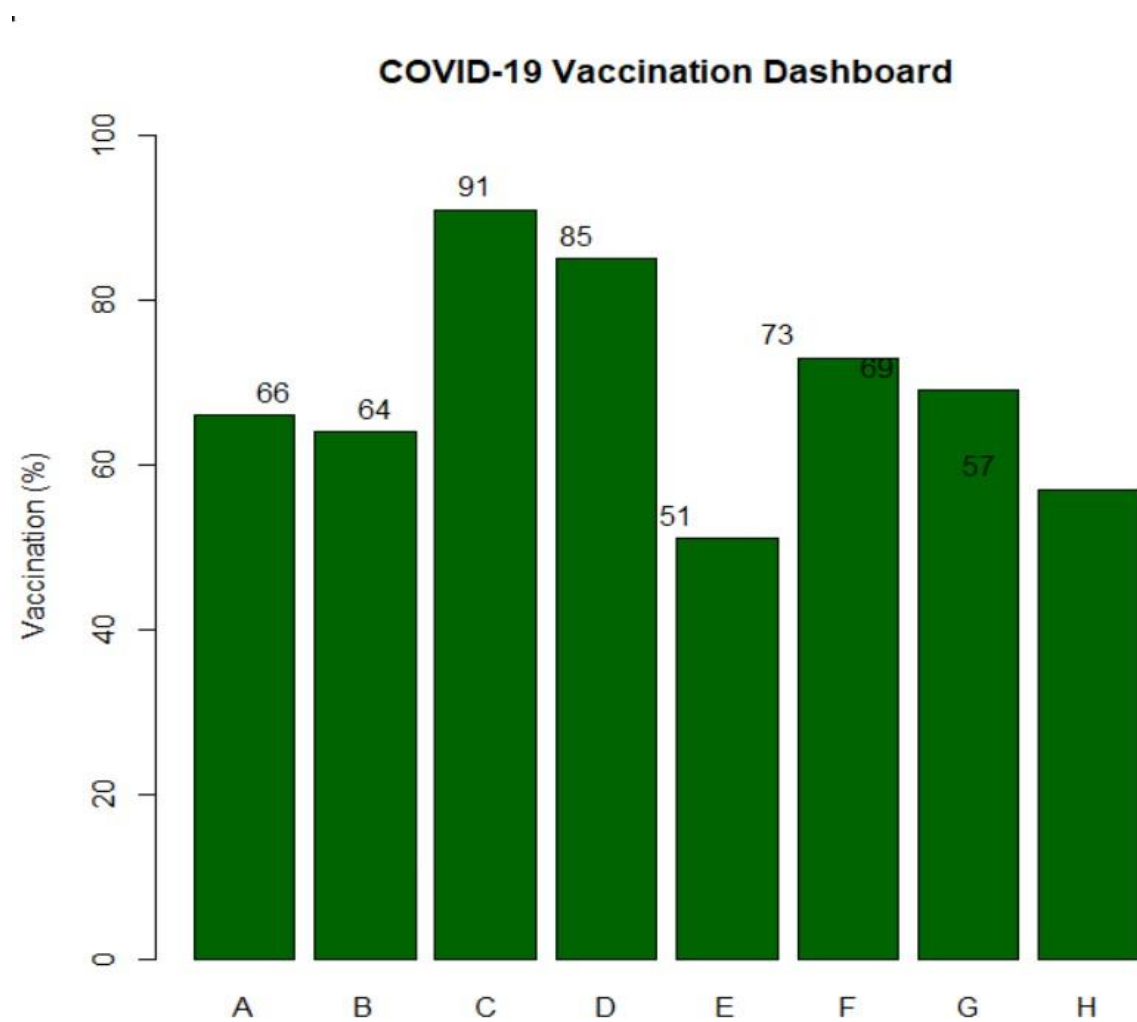


CODE: R Program

Scenario

```
districts <- c("A","B","C","D","E","F","G","H") vaccinated
<- sample(50:100,8)
barplot(vaccinated,
names.arg=districts,          col="darkgreen",
ylim=c(0,100),              main="COVID-19
Vaccination Dashboard",
ylab="Vaccination (%)")
text(1:8,
vaccinated+3,
labels=vaccinated)
```

OUTPUT:



Scenario